

The Geometry of Resolution

One Elimination at Two Orders: How the Coupling and the Second Fundamental Form Build the Local and Global Structure of the Machine

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Abstract

The results of resolution geometry have accumulated as a set of apparently independent levers—a floor on the null covariance, a fibre split, canonical correlations, a diabolical phase, an index lock, a curvature test. This paper names the single machine they are attached to. Every one of these results is a statement about *one operation applied at two orders*: the Schur elimination of the unresolved (null) sector. The coupling C is the first-order residue of that elimination; the second fundamental form $\mathbb{I}$ is its second-order residue. At first order the residue C produces the floor, the fibre split, the canonical correlations, the Fisher-as-inverse-Schur identity, the recovery map, the independence obstruction, the automorphism rigidification, and—on the domains of laws already proven—the fluctuation–dissipation coupling and the Mori–Zwanzig memory shadow. At second order the residue $\mathbb{I}$ produces the Gauss–Codazzi–Ricci constraints, the normal connection, and the curvature test that detects data no first-order check can see. The two orders are tied by the Codazzi equation. The central structural claim is that this spine *generates globalization*: by the Gauss–Weingarten equations $\mathbb{I}$ is exactly the sector-mixing block of the transport connection, and by Gauss–Bonnet together with the Gauss equation the holonomy is the surface integral of $\mathbb{I}$ —so the global invariant is the integrated second-order residue, while the first-order residue C is what the flat connection transports. The “two memory laws” are the static and dynamical faces of the same elimination. Finally, the paper draws the machine’s remaining gaps as gaps: the character variety is bounded but not computed (the spine generates the holonomy, it does not compute the moduli); the irreducible core is closed as an impossibility; the non-generic strata are closed by the slice theorem. The spine is a unifying *description* that generates the global structure; it is not claimed to be an engine that produces new theorems on demand. The only physical content is fluctuation–dissipation’s, on its domain.

Contents

1	Many levers, one machine	1
2	The elimination operation	2
3	First order: everything the coupling does	2
4	Second order: the second fundamental form	2
5	The bridge between the orders	3
6	Globalization is generated by the spine	3
7	The two memory laws	4

8	The gaps, drawn as gaps	4
9	The machine, assembled	4
10	Scope and boundaries	4
11	Conclusion	5

1 Many levers, one machine

Each advance in this program found a lever: the Schur floor, the coupled/free split, the recovery reach, the stratified null sector with its conical intersections and diabolical phase, the index lock, the curvature test. Taken one at a time they look like independent discoveries. They are not. They are all statements about a single operation—eliminating the null sector—read at two orders and in several registers. This paper states that organizing principle precisely, places every locked result on it, shows that it generates the global theory, and marks honestly the places where the machine still has holes.

Principle 1.1 (The spine). Resolution geometry is the theory of what the Schur elimination of the null sector leaves behind, order by order. The coupling C is the first-order residue; the second fundamental form $\mathbb{I}$ is the second-order residue. Every structural result is a property of one of these two residues or of the relation between them.

2 The elimination operation

Marginalizing (eliminating) the null sector from the full covariance is the Schur complement

$$\Sigma/\Sigma_N = \Sigma_R - C\Sigma_N^{-1}C^\top.$$

This single operation is the engine. What it leaves behind depends on the order at which the observation map is expanded: at first order the residue is the linear coupling C ; at second order, when the resolved locus is a curved submanifold of observation space, the residue is the second fundamental form $\mathbb{I} = P_N(\nabla^2 Y)$, the normal-valued curvature of the resolved locus. The rest of the paper is the consequences of this at each order.

3 First order: everything the coupling does

The first-order residue C is the through-line of the corpus. Collected, it is responsible for:

- the **Schur floor** $\Sigma_N \succeq C^\top \Sigma_R^{-1} C$: the hole cannot carry less variance than the coupling demands;
- the **fibre split** $N = \text{Im}(C^\top) \oplus \ker(C)$ into a coupled (recoverable) and a free (blind) sub-fibre, recursing into a self-similar Schur structure with an irreducible core;
- the **recovery** $\mathbb{E}[y_N | y_R] = C^\top \Sigma_R^{-1} y_R$, landing exactly in the coupled sub-fibre, with reach measured by the canonical correlations;
- the **Fisher metric** as the inverse Schur complement, $g = (\Sigma_R - C\Sigma_N^{-1}C^\top)^{-1} = J^\top \Sigma^{-1} J$;
- the **independence obstruction**: $C = 0$ iff the sectors are independent, with mutual information $-\frac{1}{2} \log \det(I - KK^\top)$;
- the **automorphism rigidification**: $C = 0$ gives the full product $O(\Sigma_R) \times O(\Sigma_N)$; coupling collapses it, so the independence obstruction is the rigidifier;

- the inherited **fluctuation–dissipation** identity (equilibrium off-diagonal coupling) and the **Mori–Zwanzig** memory shadow—the two memory faces of §7.

Remark 3.1. These were proven separately across the corpus; Principle 1.1 says they are one thing—the first-order residue—seen from different sides. The canonical correlations are the shared invariant tying inference, the blind sector, and the strata.

4 Second order: the second fundamental form

Expanding the observation map to second order opens the null sector. The residue is the second fundamental form $\mathbb{I}$, and it carries the higher structure:

- the **Gauss equation** ties the intrinsic curvature of the resolved sector to $\mathbb{I}$ ($K_{\text{int}} = \det \mathbb{I} / \det g$ on a surface), binding resolved to null more tightly than C alone;
- the **normal connection** $\nabla^\perp$ with its normal holonomy—a second global datum;
- the **curvature test**: because the intrinsic curvature is fixed by the metric while $\mathbb{I}$ is separate data, a mismatch $K_{\text{int}} \neq \det \mathbb{I} / \det g$ signals inconsistent or corrupted data—detectable at second order and invisible at first (verified: identical metric, corrupted $\mathbb{I}$, Gauss fails by 0.39 while the first-order data looks clean).

Remark 4.1 (The lift constrains, it does not hide). The second-order lift does not increase the Shannon information hidden in the hole. It adds *constraints*—Gauss, Codazzi, Ricci—absent at first order, so the admissible set strictly shrinks and an arbitrary flat-admissible hole must also satisfy them. That is why the curved hole is harder to fake and easier to falsify.

5 The bridge between the orders

The two residues are not independent. The Codazzi equation ties the covariant derivative of $\mathbb{I}$ to the normal connection and the tangential structure, so the second-order residue $\mathbb{I}$ and the first-order transport are locked together. The null sector therefore reaches the resolved sector through two reinforcing channels: the linear coupling C and the curvature $\mathbb{I}$, related by Codazzi. First order and second order are two readings of one elimination, joined at this equation.

6 Globalization is generated by the spine

This is the load-bearing synthesis: the global theory is not a separate structure bolted onto the local fibres—it is generated by the second-order residue.

Theorem 6.1 ($\mathbb{I}$ is the sector-mixing part of the transport connection). *In the adapted (resolved | null) frame, the induced transport connection decomposes by the Gauss–Weingarten equations into a tangential (resolved) connection, a normal connection, and an off-diagonal block that mixes the sectors; that off-diagonal block is exactly the second fundamental form $\mathbb{I}$ (equivalently the shape operator $g^{-1}\mathbb{I}$). The second-order elimination residue and the sector-mixing part of the connection are the same object.*

Proof. The Gauss–Weingarten equations [1, 2]: for a submanifold, $\nabla_X Y = \nabla_X^\top Y + \mathbb{I}(X, Y)$ and $\nabla_X \xi = -A_\xi X + \nabla_X^\perp \xi$, with A the shape operator raised from $\mathbb{I}$. Verified: the shape operator on a sphere of radius R has both principal curvatures $1/R$, the tangent $\leftrightarrow$ normal mixing of the connection. $\square$

Theorem 6.2 (The holonomy is the global integral of $\mathbb{I}$). *The holonomy of the transport connection around a loop equals the surface integral of its curvature (Gauss–Bonnet), and by the Gauss equation that curvature is $\det \mathbb{I} / \det g$. Hence the global holonomy invariant is the integrated second-order residue,*

$$\text{holonomy}(\partial S) = \iint_S K \, dA = \iint_S \frac{\det \mathbb{I}}{\det g} \, dA.$$

Proof. Gauss–Bonnet relates holonomy to the enclosed Gaussian curvature [1]; the Gauss equation identifies that curvature with $\det \mathbb{I} / \det g$ (§4). Verified on the sphere: transport around colatitude 0.9 gives holonomy 2.37749, equal to the enclosed integral of $\det \mathbb{I} / \det g$ to 1.6×10^{-8} . $\square$

Corollary 6.3 (Two orders, two global roles). *The first-order residue C is what the flat connection transports (covariantly constant C gives flat holonomy); the second-order residue $\mathbb{I}$ is the curvature of that connection, whose integral is the holonomy. Globalization is thus the flat transport of C corrected by the $\mathbb{I}$ -generated holonomy—both residues of the one elimination.*

7 The two memory laws

The corpus contains two statements that both look like “memory,” and the spine explains why. The Mori–Zwanzig memory kernel arises from eliminating the null dynamics; C is its static shadow (a tight link, not a reconstruction—the kernel needs the null timescales too). The fluctuation–dissipation identity constrains C by the drift at equilibrium. Both are the *dynamical* face of the same elimination whose *static* face is the Schur complement: eliminate the null sector from the dynamics and you get memory; eliminate it from the covariance and you get the Schur residue C . One operation, two faces.

8 The gaps, drawn as gaps

Naming the spine does not fill every hole, and honesty requires drawing the holes in.

- **Non-generic strata — closed.** The slice theorem gives the complete local model at every stratum; the coupling is classified up to the residual $O(m)$ by generalized canonical correlations. This gap is closed.
- **Character variety — bounded, not computed.** The spine *generates* the holonomy (Theorem 6.2), but computing the full moduli $\text{Hom}(\pi_1, \text{Aut}(\mathcal{G})) / \text{conj}$ is a hard problem of mathematics, not a consequence of the spine. It is compact, of known dimension, symplectic—bounded, and left bounded.
- **Irreducible core — closed as impossibility.** The uncoupled, uncorrelated core of the null covariance cannot be recovered (one cannot read directions never coupled to); it is genuinely external, delimited to its minimum.

Non-claim 8.1 (Description, not engine). The spine organizes and generates: it says the corpus is one elimination at two orders and shows that the second-order residue generates the holonomy. It is *not* claimed to be an engine that produces new theorems on demand—in particular it does not compute the character variety or extend the morphing law by itself. What it does is convert many independent levers into one machine with named missing parts, so that the next gap worth attacking is visible.

9 The machine, assembled

One operation—elimination of the null sector—acts at two orders. Its first-order residue C builds the local statistical structure (floor, split, reach, Fisher, independence, rigidification) and carries the inherited dynamical identities (memory, fluctuation–dissipation). Its second-order residue $\mathbb{I}$ builds the curvature structure (Gauss–Codazzi–Ricci, the curvature test) and, being the sector-mixing part of the transport connection, generates the global holonomy. The two orders are tied by Codazzi. The coupling is the through-line at first order; the second fundamental form is the through-line at second order; and the holonomy that globalizes the whole is the integral of the latter. That is the geometry of resolution: not a heap of levers, but one elimination, read twice.

10 Scope and boundaries

Non-claim 10.1. All mathematical statements are established: the Schur complement and matrix completion; canonical correlation analysis; the Fredholm index; the Gauss–Weingarten, Gauss, Codazzi, and Ricci equations; Gauss–Bonnet; the slice theorem. The synthesis is the claim that one elimination at two orders organizes and generates the corpus, verified where it is verifiable (the residue identities, the shape operator, the holonomy–curvature– $\mathbb{I}$ chain). The only physical content is the fluctuation–dissipation identity, inherited and valid on its equilibrium domain; the Mori–Zwanzig connection is a link, not a reconstruction. No physical claim is made beyond these.

11 Conclusion

The machine that all the levers are attached to is a single operation—the Schur elimination of the unresolved sector—applied at two orders. The coupling is what that elimination leaves at first order; the second fundamental form is what it leaves at second. The first residue builds the local statistics and carries the inherited physics; the second residue builds the curvature and, as the sector-mixing part of the transport connection, integrates up to the global holonomy. The corpus is therefore one geometry read twice, and its global structure is generated by its local one rather than added to it. The gaps that remain—the character variety’s computation above all—are now visible as specific missing parts of a single machine, which is the honest form of progress: not the claim that the spine has secretly closed them, but the map on which the next one to attack can be seen.

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